

Lesson 01 Preparation: Squares and Cubes

Getting ready for *Square and Cube Roots*

Name: _____

Date: _____

Purpose and time

Why: Next lesson you will learn to *undo* squaring and cubing. To do that quickly, you need to know your square and cube numbers.

Time: about 10–15 minutes. Work on your own. You do **not** need a calculator, a website or anyone's help.

Bring: this sheet, completed as far as you can, to the lesson.

If you get stuck

1. Read the **Remember** box and the **worked example** again.
2. Write the multiplication out in full, e.g. $4^3 = 4 \times 4 \times 4$.
3. Try a smaller number first, then come back.
4. Still stuck? Write what you tried and move on. Tick "I need help" in the self-check at the end. That is useful information, not a failure.

Remember

Square number: $n^2 = n \times n$

$$3^2 = 3 \times 3 = 9 \quad (\text{not } 3 \times 2)$$

Area of a square = side $\times$ side = side²

negative $\times$ negative = positive, so $(-3)^2 = 9$

Cube number: $n^3 = n \times n \times n$

$$2^3 = 2 \times 2 \times 2 = 8 \quad (\text{not } 2 \times 3)$$

Volume of a cube = edge $\times$ edge $\times$ edge = edge³

$$(-2)^3 = (-2) \times (-2) \times (-2) = -8$$

Part A: Retrieval (about 3 minutes)

P1. Work out: (a) 7×7 (b) 12×12 (c) $3 \times 3 \times 3$

P2. Write $5 \times 5 \times 5$ using index notation. Then work out its value.

P3. Omar says $5^2 = 10$. Is he right? Explain in one sentence.

P4. A square has side length 6 cm. Work out its area. Include units.

Part B: Worked example (read carefully, about 3 minutes)

Worked example: a cube with edge 4 cm

Question: Work out 4^3 . Then find the volume of a cube with edges of length 4 cm.

Step 1 Write the power as a multiplication: $4^3 = 4 \times 4 \times 4$. *The small 3 tells us how many 4s to multiply.*

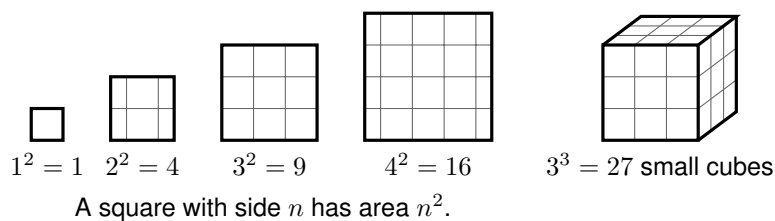
Step 2 Multiply the first pair: $4 \times 4 = 16$.

Step 3 Multiply by the last 4: $16 \times 4 = 64$. So $4^3 = 64$.

Step 4 Volume of a cube = $\text{edge}^3 = 4^3 = 64$. Volume is measured in cubic units, so the answer is 64 cm^3 .

Check 64 is *not* $4 \times 3 = 12$. A $4 \times 4 \times 4$ cube is built from 64 small unit cubes: 4 layers of 16.

Picture it: squares and cubes



Part C: Try it (about 5 minutes)

T1. Complete the table. The first two rows are done for you.

n	n^2 (square)	n^3 (cube)
1	1	1
2	4	8
3		
4		
5		
6		
10		

T2. Work out: (a) 9^2 (b) $(-4)^2$ (c) $(-1)^3$

T3. Which is bigger, 4^3 or 8^2 ? Show your working.

Part D: Notice and wonder (about 2 minutes)

- N1.** (a) A square has an area of 49 cm^2 . What is the length of one side? How did you work it out?
 (b) Another square has an area of 50 cm^2 . Can you find its side length exactly? Between which two whole numbers do you think it lies? Write what you notice or wonder.

Self-check

	I can do this	I need help with this
Work out square numbers such as 7^2 and 12^2	☐	☐
Work out cube numbers such as 2^3 and 4^3	☐	☐
Use $\text{area} = \text{side}^2$ and $\text{volume} = \text{edge}^3$	☐	☐
Square or cube a negative number	☐	☐

My question for class: